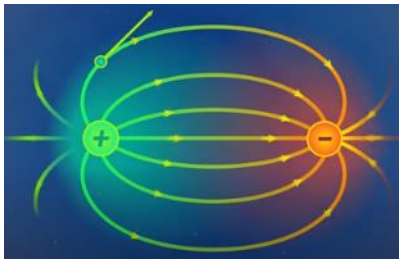
EE4047 – High Voltage Techniques I

Continuation to:
Electrostatic Fields and Planar Electrode Systems



Week 6.2 – Thursday Lesson

Asst. Prof. Dr. M. Alparslan Zehir

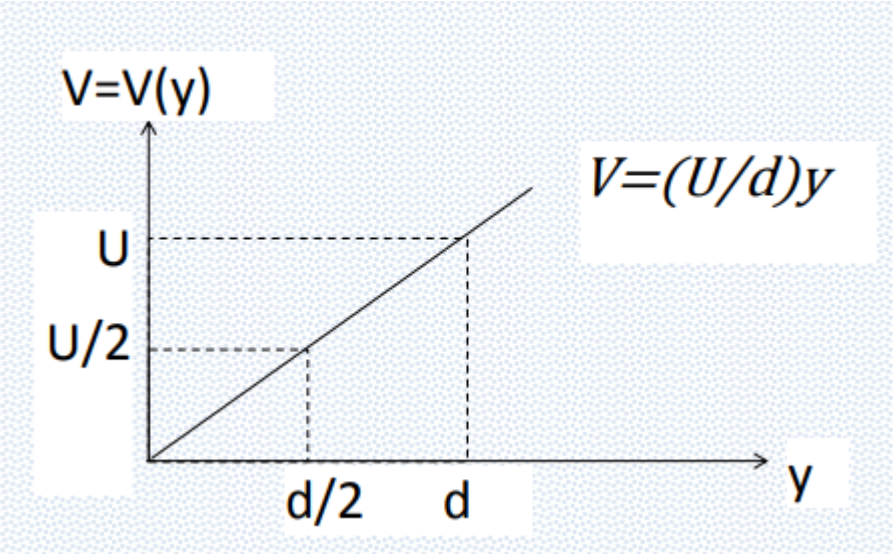
06.04.2023

OUTLINE

2 Continuation to: Electrostatic Field in Planar Electrode Systems

ELECTROSTATIC FIELD IN PLANAR ELECTRODE SYSTEMS

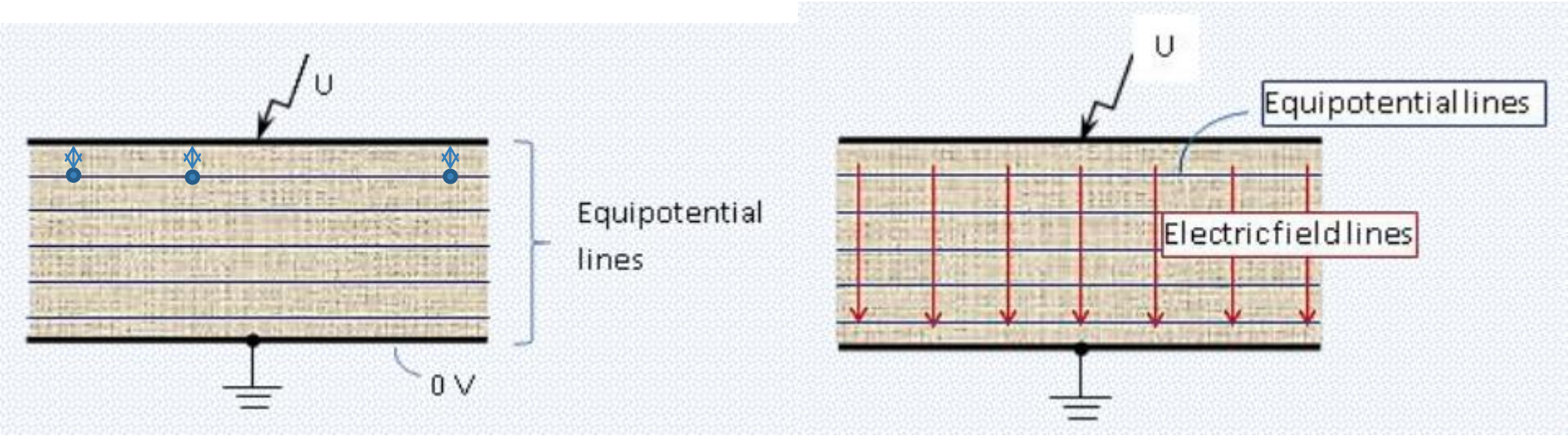
Change of voltage with distance (d):



2 ELECTROSTATIC FIELD IN PLANAR ELECTRODE SYSTEMS

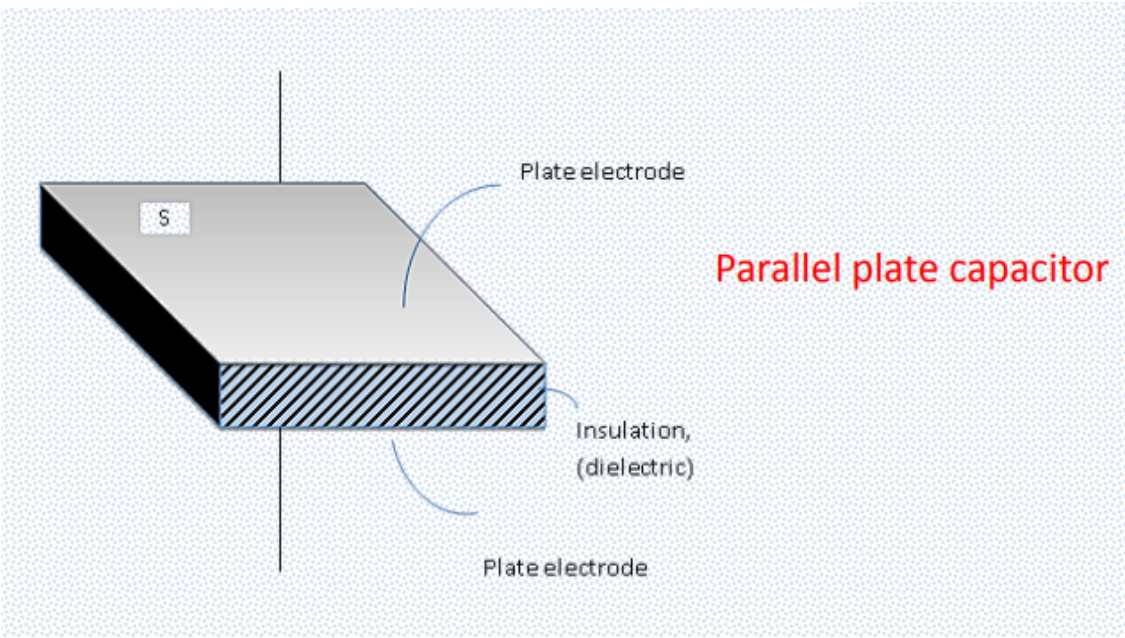
Equipotential lines, the lines on which every point has the same potential.

Electric field lines are **ALWAYS perpendicular to** equipotential lines.



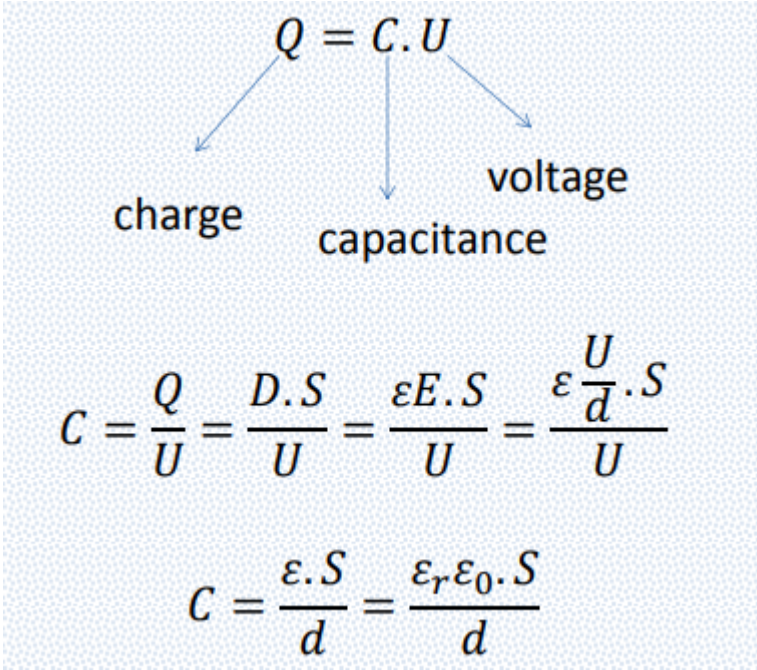
ELECTROSTATIC FIELD IN PLANAR ELECTRODE SYSTEMS

Capacitance



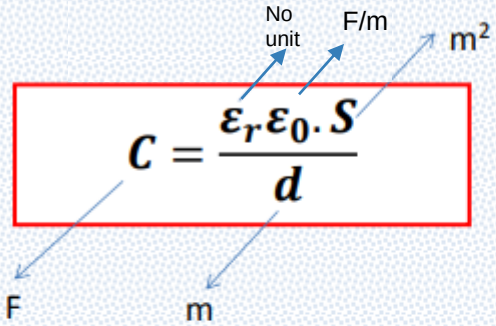
ELECTROSTATIC FIELD IN PLANAR ELECTRODE SYSTEMS

Capacitance



ELECTROSTATIC FIELD IN PLANAR ELECTRODE SYSTEMS

Capacitance



ε : Dielectric constant of the insulation = $\varepsilon_0 \varepsilon_r$ [F/m]

ε_r : relative dielectric constant (no unit), relative permittivity

$\varepsilon_0 = 8.854\ 10^{-12}$ F/m dielectric constant for space, permittivity

S : Area of electrode surface [m^2]

d : electrode separation, insulation thickness [m]

The reactance of a system to current going through because of its capacitance.

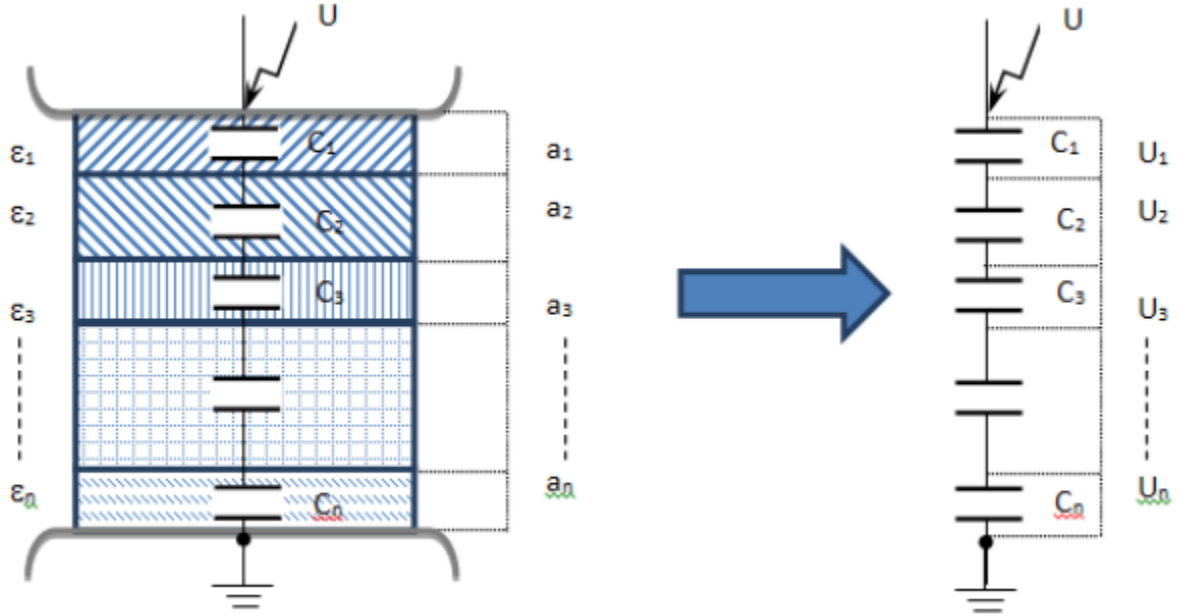
$$X_c = \frac{1}{\omega C} = \frac{1}{2\pi f\ C}$$

$$\rightarrow I = \frac{U}{R} \rightarrow I = \frac{U}{X_c}$$

$$C \uparrow \Rightarrow X_c \downarrow \Rightarrow I \uparrow$$
$$C \downarrow \Rightarrow X_c \uparrow \Rightarrow I \downarrow$$

ELECTROSTATIC FIELD IN PLANAR ELECTRODE SYSTEMS

Multilayer dielectrics in planar electrode systems



This dielectric configuration is connected in **SERIES**

$$I = I_1 = I_2 = I_3 = \cdots = I_n$$

$$Q = Q_1 = Q_2 = Q_3 = \cdots = Q_n$$

$$C\,U = C_1U_1 = C_2U_2 = C_3U_3 = \cdots = C_nU_n$$

$$U = U_1 + U_2 + U_3 + \cdots + U_n$$

U is the voltage applied to the entire system

U_1, U_2, ..., U_n are voltages across the corresponding dielectric layers

Voltage across each dielectric layer:

$$C\,U = C_1U_1 = C_2U_2 = C_3U_3 = \cdots = C_nU_n$$

$$U_1 = \frac{C\,U}{C_1}$$

$$U_2 = \frac{C\,U}{C_2}$$

$$\vdots$$

$$U_n = \frac{C\,U}{C_n}$$

Equivalent capacitance of the system:

$$X_C = X_{C1} + X_{C2} + X_{C3} + \cdots + X_{Cn}$$

$$\frac{1}{\omega C} = \frac{1}{\omega C_1} + \frac{1}{\omega C_2} + \frac{1}{\omega C_3} + \cdots + \frac{1}{\omega C_n}$$

$$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \cdots + \frac{1}{C_n}$$

C : equivalent capacitance of the system, total capacitance of the system
U : voltage applied to the entire system
C_i : capacitance of the ith dielectric layer
U_i : voltage across i^{th} dielectric layer

ELECTROSTATIC FIELD IN PLANAR ELECTRODE SYSTEMS

Calculation of capacitance:

$$C = \frac{\varepsilon . S}{d}, C_1 = \frac{\varepsilon_1 . S}{d_1}, C_2 = \frac{\varepsilon_2 . S}{d_2}, \dots, C_n = \frac{\varepsilon_n . S}{d_n}$$

$$\frac{1}{C} = \frac{1}{\frac{\varepsilon_1 . S}{d_1}} + \frac{1}{\frac{\varepsilon_2 . S}{d_2}} + \cdots + \frac{1}{\frac{\varepsilon_n . S}{d_n}} = \frac{1}{S}\left(\frac{d_1}{\varepsilon_1} + \frac{d_2}{\varepsilon_2} + \cdots + \frac{d_n}{\varepsilon_n}\right)$$

$$= \frac{1}{S}\underbrace{\sum_{i=1}^{n} \frac{d_i}{\varepsilon_i}}_{\text{A}}$$

ELECTROSTATIC FIELD IN PLANAR ELECTRODE SYSTEMS

Total capacitance, equivalent capacitance:

$$A = \sum_{i=1}^{n} \frac{d_i}{\varepsilon_i} = \frac{d_1}{\varepsilon_1} + \frac{d_2}{\varepsilon_2} + \cdots + \frac{d_n}{\varepsilon_n}$$

$$\frac{1}{C} = \frac{A}{S}$$

$$\Rightarrow C = \frac{S}{A}$$

ELECTROSTATIC FIELD IN PLANAR ELECTRODE SYSTEMS

Potential and electric field across 1st dielectric layer:

$$U_1 = \frac{C\,U}{C_1} = \frac{\frac{S}{A}U}{\frac{\varepsilon_1.S}{d_1}} = \frac{U}{A}\frac{d_1}{\varepsilon_1}$$

$$E_1 = \frac{U_1}{d_1} = \frac{\frac{U}{A}\frac{d_1}{\varepsilon_1}}{d_1} = \frac{U}{A.\varepsilon_1}$$

Potential and electric field across 2nd dielectric layer:

$$U_2 = \frac{C\,U}{C_2} = \frac{\frac{S}{A}U}{\frac{\varepsilon_2.S}{d_2}} = \frac{U}{A}\frac{d_2}{\varepsilon_2}$$

$$E_2 = \frac{U_2}{d_2} = \frac{\frac{U}{A}\frac{\varepsilon_2}{d_2}}{d_2} = \frac{U}{A.\varepsilon_2}$$

Potential and electric field across ith dielectric layer:

$$U_i = \frac{C\,U}{C_i} = \frac{\frac{S}{A}U}{\frac{\varepsilon_i . S}{d_i}} = \frac{U}{A}\frac{d_i}{\varepsilon_i}$$

$$E_i = \frac{U_i}{d_i} = \frac{\frac{U}{A}\frac{\varepsilon_i}{d_i}}{d_i} = \frac{U}{A.\varepsilon_i}$$

If electric field strength across any dielectric layer E_i, becomes greater than breakdown strength of that dielectric, there will be an electrical discharge.

$E_i \geq E_{bi}$ → Discharge in that layer of the insulation!!!

Breakdown, short circuit in i^{th} dielectric layer

After that layer is lost, the equivalent capacitance will change, the voltage across each layer will change and also new electric field strength across each dielectric layer will change too. They should be calculated based on the **remaining** dielectric layers.

Based on the breakdown strength of each layer, another layer may breakdown too or all the layers may withstand the new case. If another layer breaksdown, the same process explained above should be repeated to see if other layers will withstand to the brand new case or not.

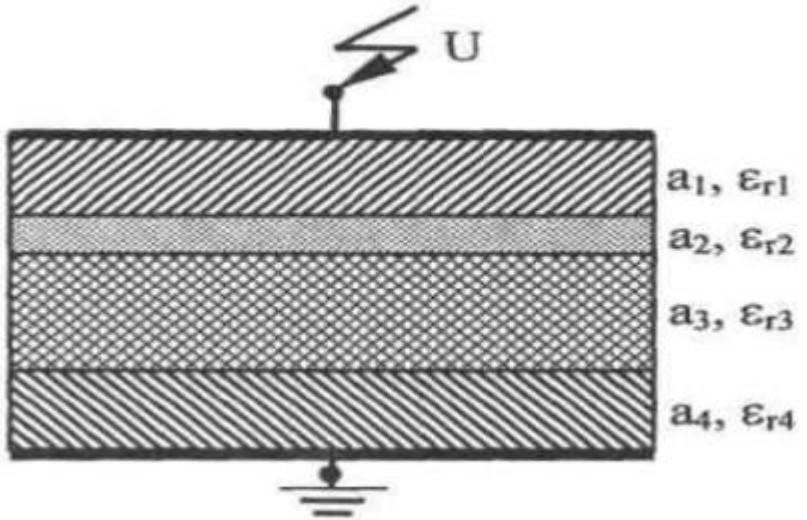
Example-1:

The 4 layer planar electrode system has U_{rms}=156 kV applied to the upper electrode. The thickness, relative dielectric constant and breakdown strength of each layer are given below:

$a_1 = 1{,}0\ cm, \quad \varepsilon_{r1} = 6, \quad E_{d1} = 80\ kV/cm$
$a_2 = 0{,}5\ cm, \quad \varepsilon_{r2} = 10 \quad E_{d2} = 55\ kV/cm$
$a_3 = 1{,}5\ cm, \quad \varepsilon_{r3} = 4, \quad E_{d3} = 100\ kV/cm$
$a_4 = 1{,}0\ cm, \quad \varepsilon_{r4} = 2, \quad E_{d4} = 60\ kV/cm$

a) Calculate the potential and electric field across each layer.

b) Find out if the system will breakdown or withstand.



Example-1:

Solution: a)The voltage across each layer in a multilayer planar electrode system:

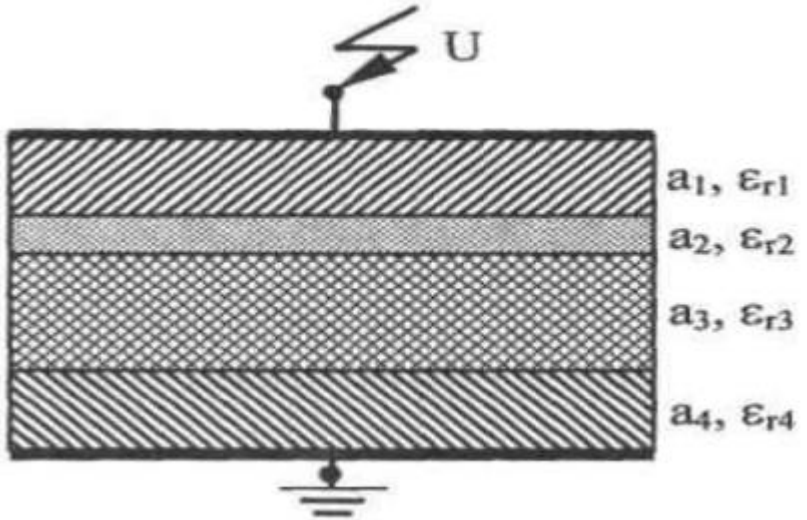
$$U_i = \frac{d_i}{\varepsilon_i}\frac{U}{A}$$

$$A = \sum_{i=1}^{4}\frac{d_i}{\varepsilon_i} = \frac{d_1}{\varepsilon_0\varepsilon_{r1}} + \frac{d_2}{\varepsilon_0\varepsilon_{r2}} + \frac{d_3}{\varepsilon_0\varepsilon_{r3}} + \frac{d_4}{\varepsilon_0\varepsilon_{r4}}$$

$$\varepsilon_i = \varepsilon_0.\varepsilon_{ri}$$

$$U_i = \frac{d_i}{\varepsilon_0.\varepsilon_{ri}}\frac{U}{\frac{1}{\varepsilon_0}\left(\frac{d_1}{\varepsilon_{r1}} + \frac{d_2}{\varepsilon_{r2}} + \frac{d_3}{\varepsilon_{r3}} + \frac{d_4}{\varepsilon_{r4}}\right)}$$

$$U = U_{rms}\sqrt{2} = 156.\sqrt{2} = 220\ kV.$$



Example-1:

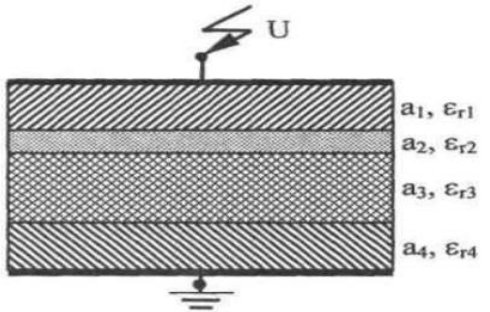
Solution: Calculation of voltages and electric field strengths:

$$U_1 = \frac{1}{6}.\frac{220}{1{,}09} = 33{,}64\ kV \quad \rightarrow \quad E_1 = \frac{U_1}{d_1} = \frac{33{,}64}{1} = 33{,}64\ kV/cm$$

$$U_2 = \frac{0{,}5}{10}.\frac{220}{1{,}09} = 10{,}09\ kV \quad \rightarrow \quad E_2 = \frac{U_2}{d_2} = \frac{10{,}09}{0{,}5} = 20{,}18\ kV/cm$$

$$U_3 = \frac{1{,}5}{4}.\frac{220}{1{,}09} = 75{,}69\ kV \quad \rightarrow \quad E_3 = \frac{U_3}{d_3} = \frac{75{,}69}{1{,}5} = 50{,}45\ kV/cm$$

$$U_4 = \frac{1}{2}.\frac{220}{1{,}09} = 100{,}91\ kV \quad \rightarrow \quad E_4 = \frac{U_4}{d_4} = \frac{100{,}91}{1} = 100{,}91\ kV/cm$$



Example-1:

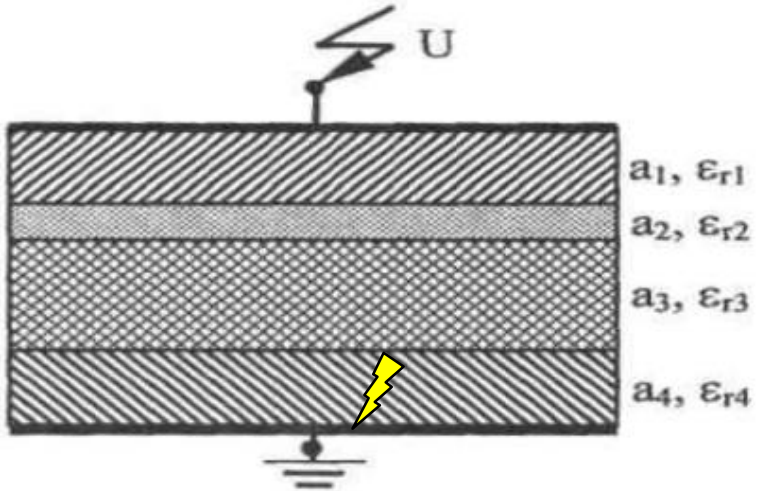
Solution: Finding out if each layer withstands or breaksdown:

$E_1 = 33{,}64\ kV/cm < E_{d1} = 80\ kV/cm$ withstands

$E_2 = 20{,}18\ kV/cm < E_{d2} = 55\ kV/cm$ withstands

$E_3 = 50{,}45\ kV/cm < E_{d3} = 100\ kV/cm$ withstands

$E_4 = 100{,}91\ kV/cm > E_{d4} = 60\ kV/cm$ **breakdown**



Example-1:

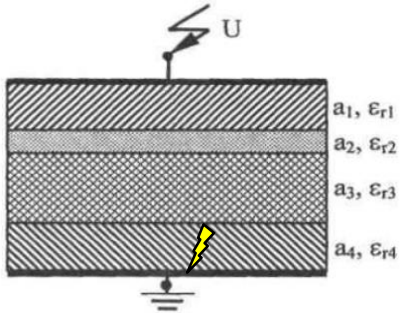
Solution: b) Because layer 4 is brokendown, the voltage will be applied to the remaining three layers.

$A_{new} = 0{,}59$

$$U_{1new} = \frac{1}{6}.\frac{220}{0{,}59} = 62{,}14\ \ kV \quad \rightarrow \quad E_{1new} = \frac{U_{1new}}{d_1} = \frac{62{,}14}{1} = 62{,}14\ kV/cm$$

$$U_{2new} = \frac{0{,}5}{10}.\frac{220}{0{,}59} = 18{,}64\ kV \quad \rightarrow \quad E_{2new} = \frac{U_{2new}}{d_2} = \frac{18{,}64}{0{,}5} = 37{,}28\ kV/cm$$

$$U_{3new} = \frac{1{,}5}{4}.\frac{220}{0{,}59} = 139{,}83\ kV \quad \rightarrow \quad E_{3new} = \frac{U_{3new}}{d_3} = \frac{139{,}83}{1{,}5} = 93{,}22\ kV/cm$$



Example-1:

Solution: b) Because layer 4 is brokendown, the voltage will be applied to the remaining three layers.

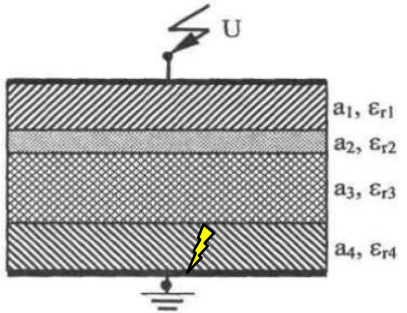
$A_{new} = 0{,}59$

$E_{1new} = 62{,}14\ kV/cm < E_{d1} = 80\ kV/cm$ withstands

$E_{2new} = 37{,}28 kV/cm < E_{d2} = 55\ kV/cm$ withstands

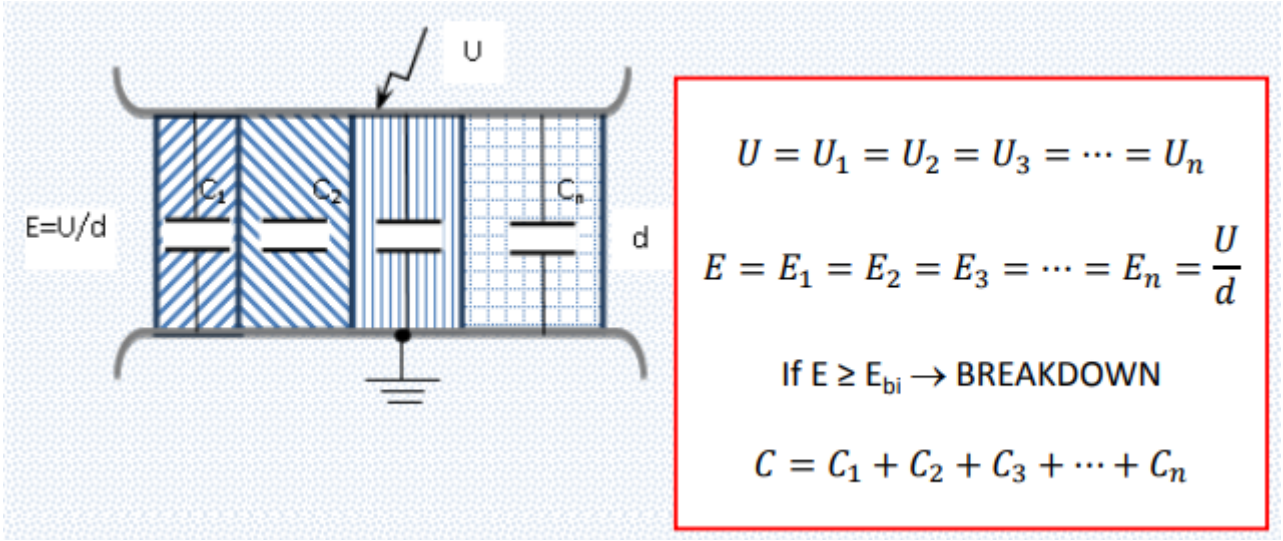
$E_{3new} = 93{,}22\ kV/cm < E_{d3} = 100\ kV/cm$ withstands

The breakdown of the 4th layer does not cause a system level breakdown The other layers still withstand to higher stress, so the system withstands to the applied voltage.



ELECTROSTATIC FIELD IN PLANAR ELECTRODE SYSTEMS

Equivalent capacitance connected in parallel



$$U = U_1 = U_2 = U_3 = \cdots = U_n$$

$$E = E_1 = E_2 = E_3 = \cdots = E_n = \frac{U}{d}$$

If $E \geq E_{bi} \rightarrow$ BREAKDOWN

$$C = C_1 + C_2 + C_3 + \cdots + C_n$$

STUDY QUESTIONS: Continuation to: Fundamentals of High Voltage Engineering

1. Please describe what is potential. What is its symbol? Draw and show on an example figure.
2. Please describe what is voltage. What is its symbol? Draw and show on an example figure.
3. What is alternating voltage? Plot an example waveform.
4. What is peak value? Show on an example figure.
5. What is rms value? Write its calculation formula and show on an example figure.
6. What is frequency?
7. What is period? Show on an example figure.
8. Why do we prefer three phase in which applications? What is neutral and ground? Draw an example figure.
9. What is an impulse voltage? Draw its waveform. Show the peak, front and tail of the impulse.

STUDY QUESTIONS: Continuation to: Fundamentals of High Voltage Engineering

10. How are the timings for lightning and switching impulse voltages? How are they calculated? Show on an example figure.

11. Describe transformers and why there are used in HV systems.

12. Describe generator types and how they are used in HV systems.

13. Describe circuit breaker types and how they are used in HV systems. Compare different circuit breaker types with each other from several aspects.

14. Describe disconnector types and how they are used in HV systems.

15. Write the names of pylon types. Draw a pylon and show its main parts.

16. Describe isolator types and how they are used in HV systems.

17. Draw a cable and show its parts. Write the different materials used in cables.

18. Why buses used in HV systems?

END OF THE LESSON

Any questions?

Check out our playlists on Youtube:

High Voltage Technique: https://www.youtube.com/playlist?list=PLLZI7WaEmZi04KMWVURGlU34W3LoNZyWX

Insulators: https://www.youtube.com/playlist?list=PLLZI7WaEmZi1ReLGq0qfP1VjRqxDJi5kA

Lightning Impulse Voltage: https://www.youtube.com/playlist?list=PLLZI7WaEmZi2isWOB-zBbb6LNplH28byQ